

Reflective Journal – Phu Nguyen

Going into this lab, I honestly didn't have a clear picture of what "types" of machine learning even meant. I just thought of ML as one big thing. But breaking it down into supervised, unsupervised, and reinforcement learning actually helped a lot. Supervised learning has the answer key (like knowing which emails are spam), unsupervised doesn't (like grouping customers without being told the groups), and reinforcement learning is its own thing where the model just learns by trying stuff and getting rewarded or punished for it, like learning to play chess by playing a ton of games.

For the wine dataset part, Logistic Regression ended up performing better than the Decision Tree. I was a little surprised because I expected the tree to do better since it feels more straightforward. But the tree was capped at a shallow depth so it couldn't get too complex, and the wine classes seem to separate pretty cleanly based on the chemical features, which probably played to Logistic Regression's strengths. I also noticed the test set was small, so I don't want to read too much into one score difference.

Overall this lab made the whole "model, evaluate, improve" cycle feel a lot less abstract than it did before.

Reflective Journal – Cyndy Ndong

I completed all the technical and written work for Lab 03 . I executed all 25 notebook cells

sequentially in Google Colab, making sure every code block ran smoothly without errors and

generated clear, accurate visual outputs. Along the way, I added inline comments to explain the

trickier parts of the code, double-checked the data preprocessing steps, and reviewed the classification metrics to evaluate how well the model performed.

Once the notebook was finished, I exported it using File then Print then Save as PDF to ensure

all the code, tables, and plots were cleanly formatted and readable. After that, I saved all my

submission files using the exact naming conventions required for the course and wrote up both

journals. Stepping through every single phase of the notebook from start to finish made sure I

covered all the assignment requirements and double-checked everything before submitting.